MRL - Milestone Review Log

Part 1 · Milestone plan · Purpose and decision standard

SIMULATED · v1.0

Study / protocol GLENOS-01 / GS-2026-01	**CIP version / date** CIP v4.2	**Site / code** Riverside Orthopedic Center / 014	**MRL plan ID** MRL-PLAN-014
Plan revision 1	**Effective date** 2 Jul 2026	**Principal Investigator** Dr Alex Morgan	**Date opened** 1 Jul 2026
Linked SRR ID / revision SRR-014 / v3 at opening; v4 linked 7 Nov	**Linked POT plan ID** POT-PLAN-014	**Plan prepared by / role** Maria Silva / Study Coordinator	**Controlled location of plan and reviews** ISF tab 3 / MRL-PLAN-014 and linked reviews

A Purpose and use

A **milestone** is a point in the investigation where a failure, if it goes unnoticed, repeats in every case that follows: the first procedures, the immediate postoperative period, the approach to endpoint assessment, close-out. At each milestone the PI takes a **gate decision**: the unit concerned (the site, a surgeon, a process or a cohort) may take its next step, may take it only within stated limits, or must wait. Participants move through the phases at different times, so a gate decides a step for its unit, not a pause for the whole study.

The MRL has two parts. Part 1 is the plan: which milestones this investigation uses, what triggers each one and what must be shown at it, approved by the PI before the first participant, with an index of every gate decision taken. Part 2 records one gate review, in a new copy each time. The MRL cites the records that hold the evidence; it does not repeat them, and its decisions do not close them.

	Part 1 · Milestone plan	Part 2 · Gate review record
Sections	A to F	I to V
Completed	Once, before the first participant. Changes go in D and apply only from their effective date.	At each gate review, in a new copy. V only when a condition or block needs clearing.
By	Record owner; approved by the PI.	Evidence reviewer prepares I to III; the PI decides in IV.
Reads	CIP, SRR and the POT plan.	The records cited as evidence: ICDL, SLCT, PFVR, FUCR, IQL, IMC, SART, CSST, PIR, CAPA log and others.

B Decision standard

Gate decision	Use when	The decision must state
Pass	The trigger was reached for the right unit and denominator, the evidence supports the next step and known limitations do not prevent it.	The scope now permitted. Pass does not mean clinical results are good.
Pass with conditions	The evidence supports the next step only within an explicit limit.	Permitted and prohibited scope; for each condition its owner, destination record, clearing evidence, re-review trigger and what happens if it is not met.
Blocked	The trigger, an essential control, an authorization or decision-critical evidence is missing, insufficient or materially uncertain.	Exactly what cannot proceed and the evidence needed to re-review. A block need not stop the whole study.

C Milestones used in this investigation

Options, not a fixed sequence: use a milestone only when its decision changes what the site may do next, and set it before the evidence exists. A calendar date alone is not a trigger.

ID	Milestone · decision it enables	What to verify · typical records	Use?	Unit	Trigger and denominator	Review window	Permitted next step
M0	**Feasibility** Site capability accepted	Operating, imaging and rehabilitation capacity; imaging gap assessment; long-term follow-up logistics. Feasibility record, SIRR.	☒ Yes ☐ No	☒ Site ☐ Surgeon ☐ Process ☐ Cohort Site 014	Before local readiness acceptance	Before start-up	Submit local feasibility evidence
M1	**Start-up** Release to enroll	Delegation with criticality codes; surgeons released and device training documented; assessors certified; instruments calibrated; imaging charter agreed; SRR open; IMP approved. RACI, CCDM, DTRM, TRE, SCR, SIRR, SRR, IMP.	☒ Yes ☐ No	☒ Site ☐ Surgeon ☐ Process ☐ Cohort Site 014	All required approvals and local releases evidenced	Before first enrollment	Enroll within existing approvals
M2	**First enrollments** Continue or expand enrollment	Consent, eligibility evidence, dual review where imaging decides, PI resolution of borderline cases, screening log. PSEAR, SIFSL.	☒ Yes ☐ No	☒ Site ☐ Surgeon ☐ Process ☐ Cohort Site 014	First 3 inclusion decisions	Before decision 4	Continue enrollment within CIP
M3	**Procedural implementation** Next procedural scope	Release current for each case, implant traceability, critical steps and substitutions documented, due learning-curve review. SCR, TRE, ICDL, SLCT.	☒ Yes ☐ No	☐ Site ☐ Surgeon ☒ Process ☐ Cohort Surgeon / process	First 3 procedures per surgeon; new operator or critical execution gap	Before next affected procedure	Continue only the cleared scope
M4	**Immediate postoperative period** Pathway continues for the cohort	Immediate review of complications and device performance; postoperative images usable. ICDL, IQL, SART, SCML.	☒ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☒ Cohort Cohort	First 3 cases with due early postoperative assessments	Before case 4; earlier for safety concern	Continue pathway within cleared scope
M5	**Approach to endpoint assessment** Endpoint assessment proceeds	Visits in window, PROMs on schedule, images usable, endpoint data agreement. PFVR, FUCR, IQL, IMC, PIR.	☒ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☒ Cohort Cohort	First 3 participants approaching the primary endpoint	At least 30 days before first endpoint visit	Proceed with endpoint assessments
M6	**Close-out** Controlled close-out	Device reconciliation signed; data completeness per participant; final safety report; archive. Accountability, IMC, PIR, SART.	☒ Yes ☐ No	☒ Site ☐ Surgeon ☐ Process ☐ Cohort Site 014	Last visit complete; close-out package ready	Before site close-out recommendation	Recommend controlled close-out
M7	**Continuous improvement** Lifecycle closure	Corrective actions closed and effective in their records; lessons in standing procedures. CAPA log, PIDLR.	☒ Yes ☐ No	☒ Site ☐ Surgeon ☐ Process ☐ Cohort Site 014	Final debrief and due effectiveness checks available	Before local lifecycle closure	Approve local lessons and handover
M_			☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			

M0 does not replace sponsor site selection or institutional approval. M1 is executable readiness, not completed forms. M3 does not certify a completed learning curve (SLCT). M5 is readiness of the assessment process, not final results. M6 leaves open cases and CAPAs in their own records. The safety stream is not a milestone: it is checked in every gate review (Part 2, III).

D Changes to the plan

A change applies from its effective date; decisions already taken stay with the plan revision they used.

Change ID	Milestone	Previous setting	New setting	Reason	Effective from	Approved by
None				No plan changes through 6 Nov 2026; revision 1 retained.		

E Plan approval

The PI approves the plan before the first participant and at each revision.

Plan revision approved	1	Approval date	1 Jul 2026

Plan prepared by

Name / role:
Maria Silva / Study Coordinator

Signature / date:
M. Silva
1 Jul 2026

Principal Investigator

Name:
Dr Alex Morgan / PI

Signature / date:
A. Morgan
1 Jul 2026

MRL - Milestone Review Log

Part 1 · Milestone plan · Gate decision index

SIMULATED · v1.0

Study / site	MRL plan ID	Index sheet no. / total	Last entry date
GLENOS-01 / Riverside 014	MRL-PLAN-014	1 of 1	6 Nov 2026

F Gate decision index

One line per signed gate decision, including re-reviews, so the sequence can be read at a glance. The detail stays in the Part 2 copy with the same Review ID.

Review ID	Milestone	Unit	Trigger met (date)	Decision date	Decision	Scope now permitted	Next milestone or re-review trigger
MRL-R01	M1 / Start-up	Site 014	3 Jul 2026	3 Jul 2026	☒ Pass ☐ With conditions ☐ Blocked	Enrollment within sponsor activation and local approvals	M2: first 3 inclusion decisions
MRL-R02	M2 / First enrollments	Site 014	31 Jul 2026	3 Aug 2026	☒ Pass ☐ With conditions ☐ Blocked	Continue enrollment within CIP	M3: first 3 procedures per surgeon
MRL-R03	M4 / Early postoperative pathway	First 3 cases	12 Aug 2026	14 Aug 2026	☒ Pass ☐ With conditions ☐ Blocked	Continue early postoperative pathway	M5: first endpoint cohort
MRL-R04	M3 / Procedural implementation	Kits and released operators	5 Nov 2026	6 Nov 2026	☐ Pass ☒ With conditions ☐ Blocked	Complete kits; currently released operators only	C-01 to C-03: verify by 15 Nov
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		

MRL - Milestone Review Log

Part 2 · Gate review record · Identification and evidence

SIMULATED · v1.0

Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total
GLENOS-01 / Riverside 014	MRL-PLAN-014 / rev. 1	MRL-R04	1 of 4

I Gate identification

Milestone (Part 1, C)	M3 / Procedural implementation	**Implementation unit**	Loaner-kit process; authorized operators
Trigger met · date	5 Nov 2026 / critical kit gap escalated	**Count reached · denominator**	8 procedures; 3 affected kit/change cases
Evidence period · cut-off	1 Jul to 5 Nov 2026 / cut-off 5 Nov	**Next step requested**	Continue unaffected procedures under restrictions
Previous decision (re-review)	Initial review of this kit restriction; R03 concerns M4	**Prepared by · date**	Maria Silva / 6 Nov 2026

II Evidence package

Cite each record so it can be found again: instrument, record ID, version or date, case or period covered, location. Adequate describes the item; it does not decide the gate.

Item	Control or element	Source · record ID · date · location	Coverage	Status	What it shows for this gate
E-01	Trigger and affected scope	PIR-014-2026-10 E-1; CAPA-014-04; 5 Nov; ISF 3	8 procedures; cases 002, 005, 007	☒ Adequate ☐ Partial ☐ Unavailable ☐ N/A	Kit/change concern documented in 3 cases; critical kit gap triggers M3.
E-02	Operative execution and kit traceability	ICDL-014-002, -005, -007 F; SLCT S-02/S-03; ISF case files	3 affected cases within 8	☐ Adequate ☒ Partial ☐ Unavailable ☐ N/A	Records support the restriction; prospective kit-check use not yet shown.
E-03	Operator release and supervision	IMC-014-02; CAPA-014-01; TRE / SCR releases; ISF training	Operators S-01 to S-03	☐ Adequate ☒ Partial ☐ Unavailable ☐ N/A	Four S-03 tasks before release in Oct; exclude any task lacking current release.
E-04	Safety and current signal review	CSST-014-02 C2/E2; SART-014-004/-005; 5 Nov; ISF safety	8 cases; 2 instability cases	☒ Adequate ☐ Partial ☐ Unavailable ☐ N/A	Safety follow-up current at cut-off; both instability cases remain in their source records.
E-05	Related data and follow-up limitations	FUCR-014-2026-10; IMC-014-02 checks 8/9; ISF 3	Oct visits: 5/7 usable; sampled fields 116/118	☒ Adequate ☐ Partial ☐ Unavailable ☐ N/A	Nurse-cover and transcription actions tracked separately; inform M5 readiness.
E-06	Accountable corrective routes	CAPA-014-01/-03/-04; POT-014-08; SRR-014; 6 Nov; ISF 3	3 active CAPAs; 04 high criticality	☐ Adequate ☒ Partial ☐ Unavailable ☐ N/A	Owners and due dates set; kit effectiveness and release audit still pending.

MRL - Milestone Review Log

Part 2 · Gate review record · Sufficiency

SIMULATED · v1.0

Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total
GLENOS-01 / Riverside 014	MRL-PLAN-014 / rev. 1	MRL-R04	2 of 4

III Sufficiency for this gate

For this milestone and unit only: judge whether the evidence in II is enough for the decision. Do not repeat the review done by the IMC or the specialist record.

1 · Trigger and scope

ID	Question	Result	Evidence item	Note
S-01	The configured trigger was reached for the right unit and the actual denominator.	☒ Complete ☐ Partial ☐ Not shown ☐ N/A	E-01	Critical kit gap reached the configured M3 trigger.
S-02	The population, evidence period and requested next step match the plan.	☒ Complete ☐ Partial ☐ Not shown ☐ N/A	E-01, E-02	8-case boundary and affected process identified.

2 · Controls and evidence

ID	Question	Result	Evidence item	Note
S-03	The critical controls for this gate were in place and their use is evidenced.	☐ Complete ☒ Partial ☐ Not shown ☐ N/A	E-02, E-06	Kit check not yet verified in prospective use.
S-04	Release, authorization, qualification and supervision are current in their records.	☐ Complete ☒ Partial ☐ Not shown ☐ N/A	E-03	Only currently released tasks may proceed.
S-05	Source records and internal verification are retrievable and enough to reconstruct what was done.	☒ Complete ☐ Partial ☐ Not shown ☐ N/A	E-01 to E-06	IDs, cut-off and locations recorded.
S-06	Required safety records, due follow-up and the latest trend review are current.	☒ Complete ☐ Partial ☐ Not shown ☐ N/A	E-04	Current at cut-off; ongoing follow-up retained.

3 · Sufficiency

ID	Question	Result	Evidence item	Note
S-07	The evidence supports the next step with no material unknown impact from a critical gap.	☐ Complete ☒ Partial ☐ Not shown ☐ N/A	E-02 to E-04	Supports restricted scope; no expansion.
S-08	Open issues have an accountable route and their limits fit the proposed scope.	☒ Complete ☐ Partial ☐ Not shown ☐ N/A	E-06	Owners, restrictions and verification dates set.

Decision-limiting gap or unknown impact

(if none, say so and cite the basis)

Prospective kit-check effectiveness and the operator-release audit are not yet demonstrated. E-02/E-03/E-06 support continued restrictions. No affected procedure may use an incomplete kit or an operator without current task release. This decision neither expands authorization nor closes CAPA or safety records.

MRL - Milestone Review Log

Part 2 · Gate review record · PI decision

SIMULATED · v1.0

Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total
GLENOS-01 / Riverside 014	MRL-PLAN-014 / rev. 1	MRL-R04	3 of 4

IV PI gate decision

Choose one decision and state what the team may now do. The decision does not close safety, risk, CAPA or authorization records.

Gate decision	☐ Pass ☒ Pass with conditions ☐ Blocked
Scope permitted from now on	Continue only procedures already permitted by the CIP and existing authorizations, using a verified complete kit and operators with current task-specific release. Record kit verification before each affected procedure.
Scope still not permitted	No procedure with missing required trial liners or unresolved kit identity. No task by S-03, or any other operator, without documented current release. No expansion of procedural scope under this decision.
PI rationale (cite II and III items)	E-01 to E-06 and S-01 to S-08 support a restricted continuation. The kit process and release checks remain incomplete; C-01/C-02 apply before each affected procedure. C-03 verifies sustained implementation. CAPA-014-04 remains high criticality and open; safety cases retain their own follow-up.
Next milestone	☐ Open ☒ Pending ☐ Blocked Objective trigger: M3: condition evidence by 15 Nov 2026; earlier for a repeat gap.

Conditions

Condition ID	Restriction	Destination record · owner	Evidence that clears it	Re-review trigger · latest date	If not met
C-01	Complete kit verified before each affected procedure	CAPA-014-04 / OR nurse lead	LS-07 rev. 2 + signed case kit check	Before affected case; review 15 Nov	Do not start with an incomplete kit
C-02	Current release checked for every assigned study task	CAPA-014-01 / Maria Silva	TRE/SCR release matched to task roster	Before assignment; review 15 Nov	Do not assign unreleased task
C-03	Retain restrictions until prospective use is verified	CAPA-014-04; POT-014-08 / PI	Next 3 affected cases: complete kit and signed checks	After 3 cases; status review 15 Nov	Keep restriction; PI re-review

Principal Investigator · decision

Name:
Dr Alex Morgan / PI

Signature / date:
A. Morgan
6 Nov 2026

Evidence reviewer

Name / role:
Maria Silva / Study Coordinator

Signature / date:
M. Silva
6 Nov 2026

MRL - Milestone Review Log

Part 2 · Gate review record · Condition clearance

SIMULATED · v1.0

Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total
GLENOS-01 / Riverside 014	MRL-PLAN-014 / rev. 1	MRL-R04	4 of 4 / update 15 Nov

V Condition clearance and re-review

Only for a condition or block. Cite the evidence actually reviewed. Clearing an MRL condition does not close its destination record; if the permitted scope changes, sign a new Part 2 decision.

Condition ID	Destination record	Evidence reviewed · period	Result	PI conclusion / new Review ID	PI initials · date
C-01	CAPA-014-04	LS-07 rev. 2 approved 12 Nov; no affected case since decision. Prospective check pending.	☐ Cleared ☒ Partly cleared ☐ Not cleared	Document updated; per-case restriction continues. No scope change.	A. Morgan / 15 Nov
C-02	CAPA-014-01	Roster reconciled with TRE/SCR on 14 Nov; S-03 excluded from unreleased tasks. Ten-task audit pending.	☐ Cleared ☒ Partly cleared ☐ Not cleared	Immediate roster control applied; retain pre-task check. CAPA remains open.	A. Morgan / 15 Nov
C-03	CAPA-014-04 / POT-014-08	0 of next 3 affected procedures available by 15 Nov. No effectiveness conclusion.	☐ Cleared ☐ Partly cleared ☒ Not cleared	Restrictions retained. Status re-review 30 Nov, earlier if evidence or repeat gap.	A. Morgan / 15 Nov
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		